

• Foundations:

- A computer learns if its Performance (P) at some task (T) improves with experience (E)

T: Task

P: Performance metrics (Accuracy, Precision, Recall, ...)

E: Experience

→ Learn from data

- Supervised learning: learn from labeled data
↳ predict new unseen data
- Unsupervised learning: learn from unlabeled data
↳ Find hidden structures
- Reinforced learning: learn by interacting
↳ maximize rewards

- How to approach:

problem definition → Collect data → Preprocess data → Feature Engineering

choose model → Train → Evaluate → Deploy → monitor

- Performance metrics (from Probability and Statistics class):

+ Accuracy: $\frac{\text{Correct}}{\text{Total}}$

+ Precision: $\frac{TP}{FP + TP}$

+ Recall: $\frac{TP}{TP + FN}$

+ F1 score: $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

for multi class:

+ confusion matrix: $C = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1k} \\ c_{21} & c_{22} & \dots & c_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ c_{k1} & c_{k2} & \dots & c_{kk} \end{bmatrix}$ with c_{ij} is num of true class j predicted as i .
Let N be $\sum_{i=1}^k \sum_{j=1}^k c_{ij}$ (total)

+ for each class: $\frac{1}{N} \sum_{k=1}^k c_{kk}$

+ Precision_k = $\frac{c_{kk}}{\sum_{i=1}^k c_{ik}}$

+ Recall_k = $\frac{c_{kk}}{\sum_{j=1}^k c_{kj}}$

+ F1_k = $2 \cdot \frac{\text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}$

for Macro Avg: $\times \frac{1}{k}$

Weighted: $w_k = \frac{N_k}{N}$

- Underfit: Train ↓ Validation ↓
- Overfit: Train ↑ Validation ↓

• Decision trees:

- A tree structure:

+ Nodes: test (eg: if else)

+ Leaves: final decision

+ Edges: outcome of Nodes

- Basis for Random Forest, easy interpretability.

- (I don't remember formula) splitting criteria:

+ Entropy: $\begin{cases} \text{high} = \text{mixed class} \\ \text{low} = \text{pure} \end{cases}$

+ Info gain: Measure entropy drop after splitting on an attribute:
to choose best split.

+ Gini: alternative impurity measure

- Mean vs median: (in class) (with example back then)

+ Mean: very sensitive to outliers

+ Median: safer when there are outliers / skew